

FORT COLLINS, CO, USA

WMO: 724697

Lat: 40.588N

Lon: 105.042W

Elev: 1505

StdP: 84.50

Time Zone: -7.00 (NAM)

Period: 86-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-19.2	-15.1	-24.7	0.5	-15.7	-20.7	0.7	-12.3	11.1	7.7	8.8	6.3	2.1	20	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	32.3	16.1	30.7	15.8	29.1	15.6	18.1	27.1	17.6	26.7	16.9	25.9	3.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.4	13.2	21.0	14.7	12.6	20.6	13.9	11.9	20.4	57.8	27.4	55.6	26.2	53.6	26.1	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.5	6.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-22.7	20.4	4.7	1.5	-26.1	21.4	-28.9	22.3	-31.5	23.2	-35.0	24.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-1.1	0.5	5.0	9.6	14.4	19.5	21.4	20.4	16.3	9.9	2.9	-2.2		
	DBStd	9.46	5.33	6.24	5.05	4.89	3.49	3.21	2.49	2.54	3.39	4.82	5.26	5.98		
	HDD10.0	1504	344	268	164	66	7	0	0	0	3	58	217	378		
	HDD18.3	3387	602	500	415	264	126	25	5	12	77	261	463	636		
	CDD10.0	1417	1	1	8	52	144	284	355	321	191	56	3	0		
	CDD18.3	257	0	0	0	1	4	59	102	75	15	1	0	0		
	CDH23.3	3144	0	0	1	49	151	731	1099	787	277	49	0	0		
	CDH26.7	993	0	0	0	7	27	262	393	252	47	5	0	0		
(14) Wind	WSAvg	2.8	2.7	2.9	3.2	3.3	3.1	2.9	2.7	2.6	2.6	2.6	2.8	2.5		
(15) Precipitation	PrecAvg	393	11	10	36	49	66	51	44	36	34	27	18	11		
	PrecMax	722	30	32	146	211	190	147	170	130	102	123	58	39		
	PrecMin	186	0	0	0	7	0	2	2	1	0	2	0	0		
	PrecStd	110	7	7	33	34	44	34	36	29	26	24	15	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	19.0	22.2	27.4	29.3	34.2	34.2	32.9	30.2	27.4	20.0	16.2		
		MCWB	5.3	7.6	8.7	12.2	13.1	16.3	15.9	15.5	14.7	11.6	8.2	6.0		
	2%	DB	12.5	14.9	19.6	24.5	27.1	31.5	32.5	31.1	28.1	24.9	17.0	12.7		
		MCWB	3.9	5.6	7.7	11.2	13.2	15.7	16.5	15.6	14.4	11.4	7.5	4.4		
	5%	DB	10.1	12.5	17.3	22.3	25.0	29.8	30.9	29.5	26.5	22.5	14.1	9.4		
		MCWB	2.8	4.4	6.7	10.5	12.7	15.2	16.4	15.9	14.0	10.1	6.1	2.3		
	10%	DB	7.6	10.0	14.6	19.4	23.0	27.9	29.1	27.6	24.8	19.7	11.6	6.5		
		MCWB	1.3	3.0	5.9	9.5	12.4	15.1	16.3	15.8	13.7	9.3	4.6	0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	8.1	9.7	13.8	15.3	18.7	19.4	18.6	17.7	13.3	9.7	6.9		
		MCDB	15.3	17.9	20.0	25.4	24.8	28.7	27.7	25.9	26.1	24.4	16.5	14.5		
	2%	WB	4.3	6.0	8.3	12.1	14.4	17.7	18.3	17.9	16.2	11.7	8.0	4.7		
		MCDB	12.1	13.9	18.6	22.0	23.2	27.3	27.7	25.7	23.9	23.4	15.8	12.2		
	5%	WB	2.9	4.6	7.2	11.0	13.7	16.8	17.6	17.4	15.2	10.6	6.5	2.6		
		MCDB	9.7	12.0	16.1	20.7	22.4	26.0	27.5	25.4	23.3	20.6	13.7	8.9		
	10%	WB	1.6	3.2	6.1	9.9	12.9	16.0	17.0	16.8	14.3	9.8	5.0	0.7		
		MCDB	7.5	9.4	14.2	19.1	21.5	25.0	26.8	24.8	22.6	19.0	11.0	6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	12.3	12.6	13.3	13.0	13.5	13.6	13.1	14.2	14.1	12.1	12.3		
		MCDBR	16.1	16.9	18.7	18.3	17.6	17.2	16.9	17.0	17.7	19.7	16.4	17.1		
		MCWBR	9.4	9.5	9.8	8.5	7.5	6.2	6.2	5.9	7.2	9.0	9.0	10.2		
		MCDBR	15.5	16.3	17.2	16.3	14.8	14.6	14.0	12.8	14.8	17.4	16.4	16.4		
	5% WB	MCWBR	9.2	9.4	9.6	8.4	6.8	6.1	6.0	5.5	6.8	8.5	9.3	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.254	0.279	0.305	0.329	0.336	0.366	0.360	0.316	0.278	0.259	0.249		
	taud		2.583	2.566	2.507	2.468	2.459	2.456	2.416	2.422	2.528	2.610	2.574	2.566		
	Ebn at Noon		948	983	982	970	946	936	904	902	928	941	919	913		
	Edh at Noon		67	80	95	106	110	111	114	110	93	76	67	63		
(44) All-Sky Solar Radiation	RadAvg		2.46	3.43	4.66	5.45	6.04	7.11	6.72	5.91	5.08	3.65	2.59	2.08		
	RadStd		0.12	0.21	0.36	0.37	0.51	0.38	0.28	0.22	0.29	0.25	0.16	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page